

UGANDA CHRISTIAN UNIVERSITY

A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY

DATA MINING, MODELING AND ANALYTICS

A consolidated empirical report on Bank Marketing and Concrete Compressive Strength

Access number: B34982

Programme: Master of Science in Data Science and Analytics

Course code: DSC8307

Academic period: Advent 2026: Year 2, Semester 1

Submission: Supplementary report accompanying executable Jupyter notebooks

Executive Summary

This report consolidates the empirical findings documented in the two supplied DSC8307 notebooks. It reports observed dataset characteristics, preprocessing decisions, benchmark results, model comparisons, diagnostic evidence, and appropriately qualified recommendations. Numerical results are transcribed from the notebook narratives and saved outputs; no new experiment is claimed in this document.

Bank Marketing: 41,188 records, 11.27% subscribers; gradient boosting ranked first on validation PR-AUC (0.4697), with held-out test PR-AUC 0.4826. A scenario-based probability threshold of 0.175 was selected under an illustrative 5:1 missed-subscriber versus unnecessary-call cost ratio.

Concrete Strength: 1,030 original observations, 1,005 after removal of 25 exact duplicates; gradient boosting achieved validation RMSE 4.688 MPa and held-out test RMSE 5.025 MPa (R^2 0.915). Residual analysis identified weaker predictions among the strongest observed concretes. Observational predictive association is not proof of a cement intervention effect.

Project 1: Bank Marketing: Classification
Course: DSC8307: Data Mining, Modeling and Analytics : Advent 2026

Research question
Which modeling approach most effectively identifies likely term-deposit subscribers, and what probability threshold balances avoidable calls against missed subscribers?

Design: stratified train/validation/test partition; preprocessing fitted on training data only; majority-class baseline; three candidate classifiers; cross-validation and randomized hyperparameter search; threshold selection on validation data; one final held-out test evaluation; sensitivity and subgroup diagnostics.

Interpretation safeguard: duration (call duration) is excluded because it is not available before the call and would leak post-contact information into a pre-call decision. Campaign variables may also depend on when a prediction is made; the notebook describes this limitation.

Sources: Moro, Cortez & Rita (2014), A Data-Driven Approach to Predict the Success of Bank Telemarketing, Decision Support Systems 62, 22–31; UCI Bank Marketing dataset; scikit-learn documentation.

Dataset exploration and audit
The original bank dataset contains 41,188 rows and 21 columns, including the binary subscription outcome. There are 4,640 positive cases (11.27%) and 36,548 negative cases. The notebook reports 12 exact duplicate rows. Several categorical predictors use the literal value “unknown” to represent unspecified information. The call-duration variable is not available before the call and is excluded from realistic pre-call prediction.

Data preparation rationale
- The target is converted to binary (yes = 1).
- Literal unknown values are treated as missing for categorical predictors, then imputed and one-hot encoded inside the fitted pipeline; this avoids treating an unspecified answer as an ordinary category without consideration.
- Numeric predictors use median imputation; numeric scaling benefits the regularized logistic model.
- Exact duplicates are reported but retained because identical records can represent different customers and no customer identifier is supplied.
- duration is excluded for pre-call use. A random stratified split estimates performance on similar records; temporal drift and repeated-customer leakage cannot be fully ruled out without identifiers and campaign dates.
- No resampling is applied initially: class imbalance is addressed using PR-AUC, recall, precision, class weights for logistic regression, and threshold choice.

Model selection: empirical answer

Three candidate classifiers were compared because they offer complementary assumptions: regularized logistic regression provides an interpretable linear log-odds benchmark, random forest accommodates nonlinear interactions, and histogram gradient boosting builds additive nonlinear corrections. Validation PR-AUC ranks gradient boosting first (0.4697), random forest second (0.4618), and logistic regression third (0.4417). The corresponding five-fold training CV means are approximately 0.4611, 0.4579, and 0.4448. Because subscriptions are uncommon (11.27%), PR-AUC is more decision-relevant than accuracy alone; the majority-class baseline achieves only approximately 0.1126 validation PR-AUC. The small gaps between the leading models do not establish statistically significant superiority. Gradient boosting is selected by the prespecified validation criterion, not by its test score. The test partition is evaluated only after selection.

Threshold decision and operational interpretation

Recommended research threshold: 0.175, chosen on the validation set under the explicitly illustrative cost assumption that a missed subscriber has five times the cost of an unnecessary call. At this cutoff, the validation sample yields 548 true positives, 733 false positives, and 380 false negatives, with 1,281 customers flagged. Thus, the policy retrieves 548 of 928 observed subscribers (recall ≈ 59.1%), while 548 of 1,281 flagged customers subscribe (precision ≈ 42.8%); 733 flagged customers do not subscribe. These are predictive classifications, not estimates of causal conversion uplift. Lower cutoffs generally increase recall and call volume; higher cutoffs reduce calls at the cost of more missed subscribers. The cost-ratio sensitivity table directly above shows that the selected cutoff changes with the relative costs. No bank-provided call costs, customer lifetime values, or capacity constraints were available; therefore 0.175 is a defensible scenario-based recommendation, not a verified profit-maximizing operating policy. Before deployment, validate probability calibration, operational costs, capacity, consent requirements and incremental campaign lift.

5. Completed interpretation, limitations and business recommendation

Class imbalance and baseline. Of 41,188 records, 4,640 are subscribers (11.27%) and 36,548 are non-subscribers. Predicting the majority class for everyone would appear accurate but would identify no prospective subscribers. Accordingly, PR-AUC and precision–recall trade-offs, not accuracy alone, guide selection.

Selected approach and evidence. Gradient boosting leads on validation PR-AUC (0.4697), narrowly ahead of random forest (0.4618) and logistic regression (0.4417). Five-fold CV provides corroborating, but not conclusive, evidence.

Held-out test PR-AUC is 0.4826, and ROC-AUC is 0.8081; these estimates should be interpreted with the reported bootstrap intervals and without claiming the small ranking differences are definitive.

Marketing decision. The validation-derived 0.175 threshold, under a hypothetical 5:1 missed-subscriber-to-unnecessary-call cost ratio, flags 1,281 validation customers: 548 true subscribers

and 733 non-subscribers, while missing 380 subscribers. The threshold illustrates a preference for recovering more potential subscribers at the expense of additional calls. The business must replace this hypothetical cost ratio with measured economics and assess incremental uplift using a randomized campaign before adopting the rule.

Diagnostics and uncertainty. The notebook reports a held-out Brier score and reliability plot to evaluate probability calibration, along with subgroup descriptive summaries and permutation importance. These diagnostics can reveal weaknesses but cannot establish fairness, explain causal mechanisms, or certify transferability. Feature importance is a predictive property, not an intervention effect. Bootstrap intervals quantify conditional test-sample uncertainty, not all uncertainty from model selection or retraining.

Data quality and deployment constraints. The literal unknown entries are treated as missing categorical responses and imputed within training-only pipelines. Twelve exact duplicate records are retained because customer identities are unavailable; this creates possible leakage across a random split. The duration field is excluded because it is unavailable before calling; campaign, contact timing, and macroeconomic fields must also be audited for availability at the exact decision point. The historical Portuguese campaign data may not generalize to future time periods or another banking population. Chronological out-of-time testing, privacy and consent review, monitoring of subgroup errors, recalibration and prospective evaluation are necessary before deployment.

Conclusion. The selected classifier materially improves subscriber prioritization over the majority baseline under this experimental design. It is suitable for further pilot evaluation, not as an automatically deployed contact policy or as proof that contacting a predicted subscriber will cause a deposit.

References

Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. Decision Support Systems, 62, 22–31. https://doi.org/10.1016/j.dss.2014.03.001

UCI Machine Learning Repository. Bank Marketing. https://archive.ics.uci.edu/dataset/222/bank+marketing scikit-learn. User Guide. https://scikit-learn.org/stable/user_guide.html

6. Final empirical synthesis and answer to the assignment

Assignment question	Completed evidence-based answer
Explore, prepare, and establish a baseline	41,188 observations, 21 original columns, 11.27% positive target; inspect `unknown` categorical values, retain 12 exact duplicate records with a documented caveat, exclude post-call `duration`, fit imputers/encoders on

	training folds only. Majority-class validation PR-AUC ≈ 0.1126.
Select and rank at most three approaches	1. Gradient boosting (validation PR-AUC 0.4697; CV mean ≈ 0.4611); 2. Random Forest (0.4618; ≈ 0.4579); 3. Logistic regression (0.4417; ≈ 0.4448). Rankings are descriptive and split-dependent.
Recommend a threshold and explain trade-offs	0.175 on validation under an illustrative 5:1 missed-subscriber-to-unnecessary-call cost ratio: TP 548, FP 733, FN 380, 1,281 flagged. Higher threshold means fewer unnecessary calls but more missed subscribers; lower threshold reverses that balance.

Independent held-out check. Gradient boosting achieved test PR-AUC 0.4826, ROC-AUC 0.8081, and accuracy 0.8744. Conditional bootstrap 95% intervals were approximately [0.452, 0.516] for PR-AUC and [0.791, 0.824] for ROC-AUC. The notebook also displays the final confusion matrix and calibration diagnostics. The test set did not determine model or threshold selection.

Decision and boundaries. Use the selected classifier as a candidate for a consent-aware pilot, with a chronological holdout, measured campaign costs, capacity constraints and a randomized uplift test. A predictive association is not evidence that a marketing call will cause a subscription. Subgroup patterns must be evaluated carefully; demographic features may act as proxies and should not be treated as a license for unfair targeting.

Selected notebook figures

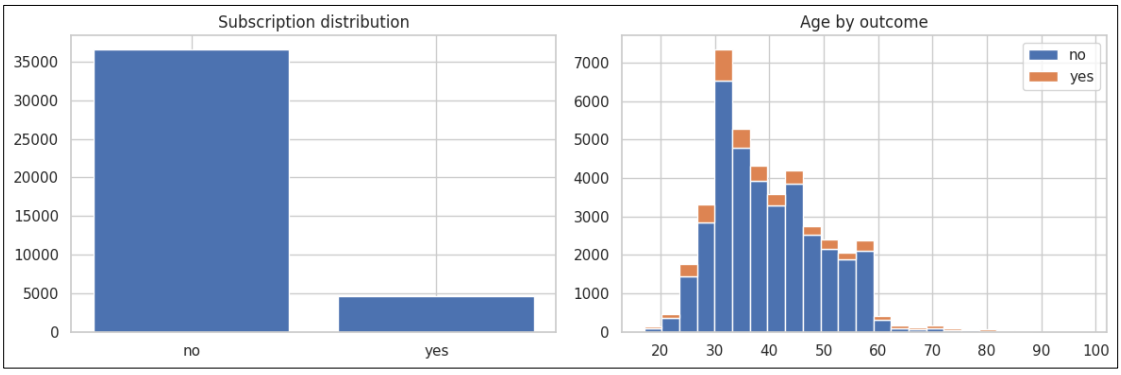


Figure 1.1. Visualization reproduced from the executed Project 1 notebook.

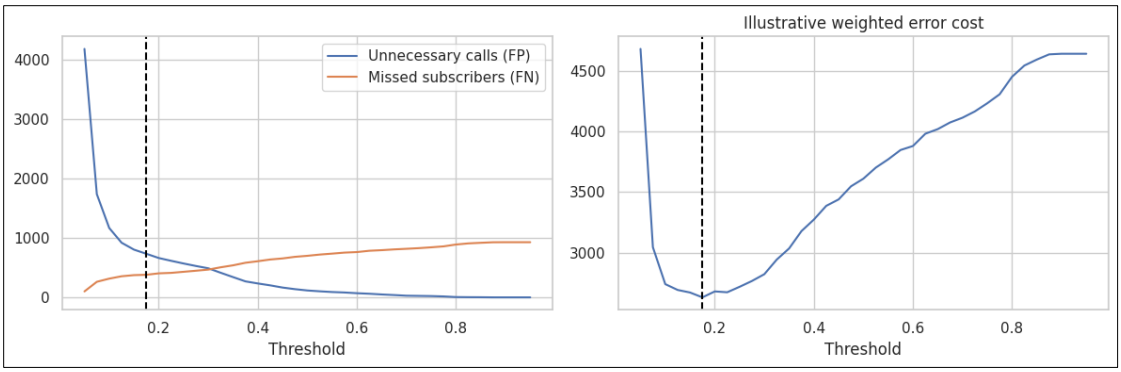


Figure 1.2. Visualization reproduced from the executed Project 1 notebook.

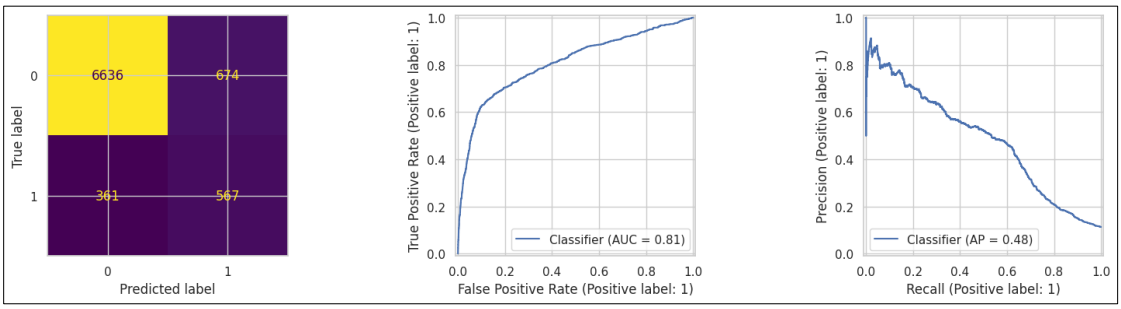


Figure 1.3. Visualization reproduced from the executed Project 1 notebook.

Project 2: Concrete Compressive Strength: Regression

Project 2: Concrete Compressive Strength Prediction

Student: Pius OGWAL | **Access number:** B34982

Course: DSC8307 — Data Mining, Modeling and Analytics | **Advent 2026**

Research question

How accurately can concrete compressive strength be predicted from ingredient proportions and curing age? Do nonlinear approaches outperform linear regression, and where do errors concentrate?

Design: training/validation/test split; training-only preprocessing; mean baseline; linear regression, polynomial ridge regression, random forest and histogram gradient boosting; cross-validation, residual and subgroup diagnostics; bootstrap uncertainty and interpretation of predictive association versus causality.

Source: Yeh, I.-C. (1998), *Modeling of strength of high-performance concrete using artificial neural networks*, Cement and Concrete Research, 28(12), 1797–1808; UCI Concrete Compressive Strength dataset.

Dataset exploration and audit

The concrete dataset contains 1,030 observations, eight numeric input variables (seven mixture ingredients and curing age), and compressive strength in MPa. No missing entries were found. The notebook identifies 25 exact duplicates and removes them before partitioning, retaining 1,005 unique records. No extreme observations are automatically removed because physically unusual mixes may still be valid.

Data preparation and assumptions

The target is the column containing "strength"; all remaining columns should be numerical mixture variables and curing age. Values are validated, missing predictors are imputed within the training pipeline, and scaling is used for regularized polynomial regression. Exact duplicate rows are inspected and **removed before splitting** to reduce identical-mixture leakage; however, repeated mixtures with different measurements can remain and limit generalization claims. A random split measures interpolation to similar mixtures, not extrapolation to unseen plants or formulations. No outliers are automatically deleted because extreme mixtures may be physically meaningful.

Completed model comparison and selection

The **mean-prediction baseline** gives validation RMSE **15.653 MPa** and R^2 **−0.015**, demonstrating that simply predicting the training mean is inadequate. The tested approaches rank by validation RMSE as follows: **1. Histogram gradient boosting (4.688 MPa; R^2 0.909), 2. Random forest (5.242 MPa; R^2 0.886), 3. Degree-two polynomial ridge (7.377 MPa; R^2 0.775), 4. Linear regression (10.114 MPa; R^2 0.576)**. Five-fold training CV RMSE means are approximately **5.176, 5.915, 8.025 and 10.421 MPa**, respectively. This is strong evidence of predictive nonlinearity in these data, but not proof of a particular physical law. Gradient boosting was selected using validation performance; the independent test set was not used for selection.

Completed residual interpretation

The validation residual is defined as **observed strength minus predicted strength**. The largest absolute validation residual is approximately **19.733 MPa**, showing that even the leading model can be substantially wrong for individual mixes. The **highest observed-strength quartile** has the largest mean absolute error (**4.057 MPa**) compared with approximately **2.727–3.641 MPa** in the other quartiles; predictions are therefore weakest, by this descriptive grouping, for the strongest observed concretes. In the **29–90-day curing-age band**, the mean signed residual is approximately **−1.956 MPa** (n=30), suggesting average overprediction in this small group. The plotted residuals versus fitted values and ingredients allow visual inspection of nonlinearity, unequal error spread and outliers; none of these diagnostics alone proves causal relationships or stable performance for unseen formulations. Because strength quartiles are defined using observed outcomes, they are retrospective diagnostics rather than deployable screening categories.

5. Does cement–strength association prove a causal effect?

Answer: No. A predictive association between cement content and compressive strength does **not** identify what would happen if cement were deliberately increased or decreased. This dataset

contains observed mixture recipes rather than randomized cement interventions. Cement content is jointly related to water, supplementary binders, aggregates, admixtures, age and processing conditions. Changing cement content may change the water–cement ratio, workability, hydration kinetics and the feasible balance of the mixture; consequently the statistical contrast between recipes is not necessarily the effect of a controlled change.

In causal notation, a model of **E[strength | cement = c, other recorded features]** does not automatically identify **E[strength | do(cement = c)]**. Unmeasured batching, temperature, compaction and curing conditions may also affect both mixture choices and outcomes. Even a high predictive R^2 or a strong feature-importance score does not establish causal identification.

Appropriate empirical test. Define a physically feasible intervention (for example, a prespecified cement replacement while controlling total binder, water–binder ratio and curing regime), randomize or systematically assign feasible mix designs, manufacture replicated batches, measure strength at standardized ages, and compare effects with uncertainty intervals. Validate any proposed formulation against applicable engineering standards. The current model can screen hypotheses for such experiments but cannot replace them.

6. Completed empirical findings, engineering interpretation and conclusion

Assignment question	Completed evidence-based answer
Explore, prepare and establish a baseline	1,030 observations; eight numerical mixture/age predictors and one strength target (MPa); no missing entries; 25 exact duplicates removed before partitioning, leaving 1,005 unique records. Mean baseline validation RMSE 15.653 MPa (R^2 −0.015).
Compare linear and nonlinear approaches	Validation RMSE: gradient boosting 4.688, random forest 5.242, polynomial ridge 7.377, linear regression 10.114 MPa. Corresponding CV RMSE means ≈ 5.176, 5.915, 8.025, 10.421 MPa. Nonlinear methods perform better on this split.
Inspect residuals and weakest predictions	Largest absolute validation residual ≈ 19.733 MPa; highest observed-strength quartile has greatest MAE (4.057 MPa). For ages 29–90 days, mean signed residual ≈ −1.956 MPa on 30 observations, indicating a tentative overprediction tendency.
Does cement association prove intervention effect?	No. Correlated mixture ingredients, composition constraints, curing conditions and unmeasured process factors prevent causal interpretation without a suitable controlled experiment.

Held-out test evidence. The validation-selected gradient boosting model achieved MAE 3.431 MPa, RMSE 5.025 MPa, and R^2 0.915 on the previously unused test partition. Conditional bootstrap 95% intervals were MAE [2.907, 3.961], RMSE [4.141, 5.840], and R^2 [0.888, 0.940].

These intervals condition on the fitted model and test-sample resampling; they do not include uncertainty from data collection or hyperparameter selection. A separate curing-age shift exercise gave RMSE ≈ 5.483 MPa and $R^2 \approx 0.794$, indicating sensitivity to the prediction population; this check is not a second unbiased test set.

Engineering implications. The error reduction over both the mean baseline and linear regression supports nonlinear prediction as a useful *screening* tool for recipes similar to those observed. However, errors of several MPa, especially for high-strength specimens, may matter for compliance or safety decisions. The model should not be used as the sole basis for accepting structural concrete. Future evaluation should include new production batches, materials and sites, age-stratified validation, uncertainty near engineering specification thresholds, and independent laboratory compression testing.

Study limitations. Random partitioning estimates interpolation to related mixtures, not transfer to new cement suppliers, aggregate sources or curing environments. Removing exact duplicates reduces, but does not eliminate, dependence between similar mixes. Observational correlations do not identify causal effects. The empirical model ranking is conditional on the provided dataset and experimental design.

Conclusion. Gradient boosting is the best-performing evaluated approach under the stated validation criterion. It generalizes well to the held-out random test split but requires external engineering validation before practical use. The answer to the causal question is unequivocally no: model predictions cannot by themselves establish what will happen after changing cement content.

References

Yeh, I.-C. (1998). Modeling of strength of high-performance concrete using artificial neural networks. *Cement and Concrete Research, 28*(12), 1797–1808. https://doi.org/10.1016/S0008-8846(98)00165-3

UCI Machine Learning Repository. *Concrete Compressive Strength. https://archive.ics.uci.edu/dataset/165/concrete+compressive+strength. scikit-learn. User Guide. https://scikit-learn.org/stable/user_guide.html

Selected notebook figures

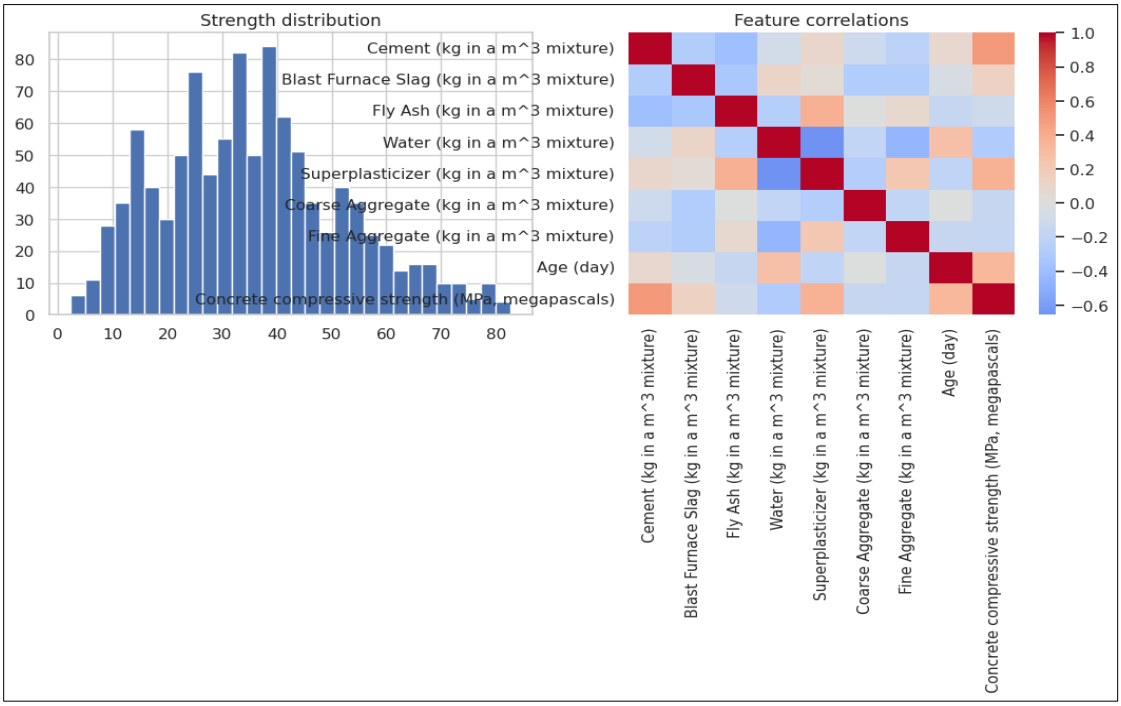


Figure 2.1. Visualization reproduced from the executed Project 2 notebook.

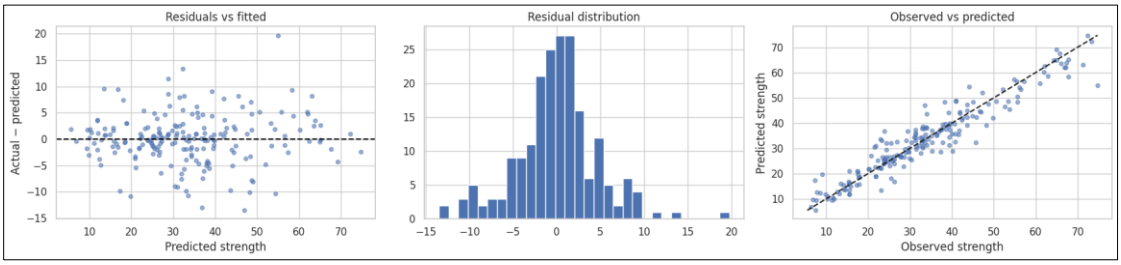


Figure 2.2. Visualization reproduced from the executed Project 2 notebook.

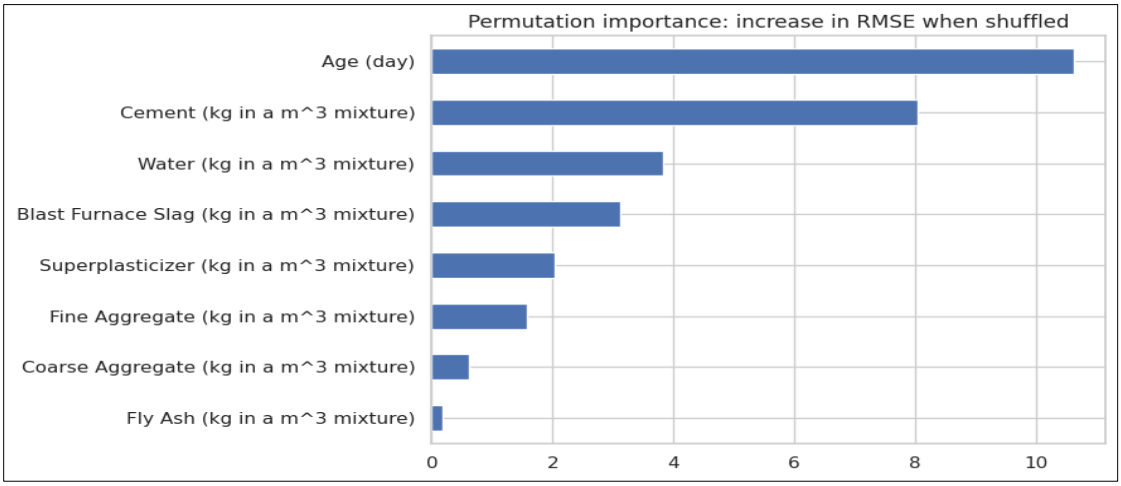


Figure 2.3. Visualization reproduced from the executed Project 2 notebook.

Submission and Reproducibility Note

This word document supplements, but does not replace, the two executable Jupyter notebooks and requirements.txt mandated by the assignment. For full traceability, readers should inspect the corresponding code cells, model outputs, plots and reproducibility instructions in the GitHub repository. The documented results arise from the supplied datasets and notebook experimental splits; they are not external-validation or causal-impact estimates.

The bank threshold depends on hypothetical operating costs; deployment requires measured costs, consent review, temporal validation and prospective uplift testing. Concrete predictions are screening aids and must not replace engineering standards or laboratory strength testing.